

PAPER_284: M16 Eagle Nebula UQFF — Dual Mass Co-Action Product (Φ_{dm})

SFR Growth $\times$ Photoevaporation Erosion Multiplicative Coupling

Classification: UQFF 2.0 Gravitational Physics — Nebular Mass Dynamics

System: M16 Eagle Nebula (IC 4703), Eagle Nebula Star-Forming Region

Session: 80 | **Module:** M16_UQFF_MODULE.cpp (22nd C++ UQFF module)

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Abstract

This paper introduces the **Dual Mass Co-Action Product** (Φ_{dm}) — a UQFF gravity modulation factor that couples star-formation-driven mass accumulation and radiation-driven photoevaporation erosion through a **multiplicative** product rather than the previously used additive form. For M16 (Eagle Nebula), with star formation rate $SFR = 1 \text{ M}_{\odot}/\text{yr}$ over initial gas mass $M_0 = 1200 \text{ M}_{\odot}$, and maximum photoevaporation fraction $E_0 = 0.3$ (30%), the multiplicative form produces a 24.3% reduction in Φ_{dm} relative to the additive approximation at $t = 5 \text{ Myr}$. This is the **first UQFF module** to simultaneously apply an additive-gain and saturation-subtractive product on the same gravity term.

2. Physical Motivation

In active star-forming nebulae, two competing processes drive mass evolution:

1. **Star Formation Accretion** — molecular gas accretes onto protostars, increasing the effective gravitational mass fraction by $SFR_rate \times t$.
2. **Photoevaporation Erosion** — UV radiation from newly formed massive stars erodes the surrounding gas, progressively reducing the effective mass by a saturating fraction $E_{rad}(t)$.

The **additive approximation** (used in prior UQFF modules):

$$\Phi_{dm}^{add} = g_{base} \times (1 + M_{sf}) - E_{rad}$$

linearly superposes the two effects, implicitly treating them as independent processes acting on separate mass reservoirs.

The **multiplicative form** (this paper):

$$\Phi_{dm}(t) = (1 + SFR_rate \times t) \times (1 - E_{rad}(t))$$

correctly encodes that **the eroded mass is drawn from the same growing reservoir** — the fraction lost to photoevaporation scales with the mass being accreted, not the original quiescent mass. This is physically accurate for pillar-geometry star formation (e.g., M16's "Pillars of Creation").

3. Mathematical Formulation

3.1 Parameters (M16 Eagle Nebula)

Parameter	Value	Description
M_0	$2.387 \times 10^{33} \text{ kg (1200 } M_\odot)$	Initial nebula gas mass
SFR	$1 M_\odot/\text{yr}$	Star formation rate
SFR_rate	$2.639 \times 10^{-11} \text{ s}^{-1}$	$= \text{SFR} / (M_0/M_\odot) / (3.156 \times 10^7 \text{ s/yr})$
τ_{erode}	$9.468 \times 10^{13} \text{ s (3 Myr)}$	Photoevaporation e-folding time
E_0	0.3	Maximum photoevaporation fraction
g_{base}	$1.454 \times 10^{-12} \text{ m/s}^2$	$G \times M / r^2$ at $r = 3.31 \times 10^{17} \text{ m}$

3.2 Dual Co-Action Product

$$M_{sf}(t) = \text{SFR_rate} \times t$$

$$E_{rad}(t) = E_0 (1 - e^{-t/\tau})$$

$$\Phi_{dm}(t) = (1 + M_{sf}) \times (1 - E_{rad})$$

3.3 Dynamic Gravity Term

$$g_{dyn}(t) = g_{base} \times \Phi_{dm}(t)$$

3.4 Multiplicative-Additive Gap

The gap relative to the additive form:

$$\Delta_{gap} = \Phi_{dm}^{mult} - \Phi_{dm}^{add} = -(M_{sf} \times E_{rad})$$

This cross-term is always **negative** — the multiplicative form predicts lower gravity than the additive approximation whenever both SFR accumulation and erosion are simultaneously active.

4. Numerical Results at $t = 5 \text{ Myr}$

$$t = 5 \text{ Myr} = 1.578 \times 10^{14} \text{ s}$$

Quantity	Value
M_sf_frac	$\text{SFR_rate} \times t = 2.639 \times 10^{-11} \times 1.578 \times 10^{14} = \mathbf{4164.8}$
E_rad	$E_0 \times (1 - \exp(-5/3)) = 0.3 \times 0.8110 = \mathbf{0.2433}$
Φ_{dm} (multiplicative)	$(1 + 4164.8) \times (1 - 0.2433) = 4165.8 \times 0.7567 = \mathbf{3151.9}$
Φ_{dm} (additive) gap_mult_add	$(1 + 4164.8) - 0.2433 = \mathbf{4165.6}$ $-(4164.8 \times 0.2433) = \mathbf{-1013.3}$ (24.3% less)
g_dyn(5 Myr)	$1.454 \times 10^{-12} \times 3151.9 = \mathbf{4.583 \times 10^{-9} \text{ m/s}^2}$

The multiplicative gap of -1013.3 confirms that treating erosion as acting on the growing mass reservoir (not the static initial mass) produces a **measurable 24.3% reduction** compared to the additive approximation.

5. Connection to UQFF 2.0 g_total

In the full M16 UQFF 2.0 equation:

$$g_{\text{total}}(r, t) = [g_{\text{dyn}}(t) + U_{g,\text{sum}}(26) + \Lambda + Q + L + F + g_{\text{exp}}] \times \text{corr}_{SC}$$

The Φ_{dm} product modulates only the dynamic base gravity term g_{dyn} . The 26-layer Triadic ($U_{g,\text{sum}}$), cosmological Λ , quantum, Lorentz, fluid, and Friedmann expansion terms are all independent of Φ_{dm} — the modulation is cleanly scoped to the time-evolving mass component.

6. UQFF Historical Distinction

Module	SFR Term	Erosion Term	Form
Session 55 CP3	$g_{\text{base}} \times (1 + M_{\text{sf}})$	$-E_{\text{rad}}$	Additive
M16EagleNebulaRadiationSFR			
This paper (PAPER_284)	$(1 + M_{\text{sf}})$	$\times (1 - E_{\text{rad}})$	Multiplicative

This is the **first UQFF module** to use the multiplicative dual co-action form, correctly encoding the coupled feedback between star formation accretion and pillar photoevaporation for M16-class nebulae.

7. Wolfram KB Term

```
M16UQFF:Phi_dm=(1+SFR_rate*t)*(1-E_0*(1-Exp[-t/tau]));      SFR_rate=2.639e-  
11/s; M_sf_frac=SFR_rate*t [PAPER_284]
```

8. Cross-References

- **PAPER_285:** Erosion Saturation Half-Time (t_{half} , Δg_{Max})
 - **PAPER_286:** Nebular Friedmann Redshift (κ_{neb} , $z=0.0015$)
 - **M16_UQFF_MODULE.cpp:** Full UQFF 2.0 C++ implementation (22nd module)
 - **CondensedPhysics3.py:** M16DualMassCoActionProductCalculator
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